ZENER DIODE

Semiconductor -1



CONCEPT
PYQs

JEE MAIN



by Mohit Goenka, IIT Kharagpur

EDUNITI

EDUNITI YouTube Channel
Learn Like Never Before

List of Content on Eduniti YouTube Channel:

1. PYQs Video Solution Topic Wise:
   (a) JEE Main 2018/2020/2021 Feb & March
2. Rank Booster Problems for JEE Main
3. Part Test Series for JEE Main
4. JEE Advanced Problem Solving Series
5. Short Concept Videos
6. Tips and Tricks Videos
7. JEE Advanced PYQs
8. Formulae Revision Series

.....and many more to come

EDUNITI

Eduniti for Physics

TOPICS COVERED

1. Fundamentals of PN Diode
2. Forward and Reverse Biased
3. Zener Diode
4. PYQs (Build your understanding)

EDUNITI

Playlist Link in Description

MODERN PHYSICS - PART 1
FORMULAE
ATOMIC PHYSICS

MODERN PHYSICS - PART 2
FORMULAE
PHOTOELECTRIC EFFECT

"FREE PDF NOTES"

EM WAVES



ELECTROSTATICS
FORMULAE

ALTERNATING CURRENT
FORMULAE

...MANY MORE

Eduniti for Physics

EDUNITI

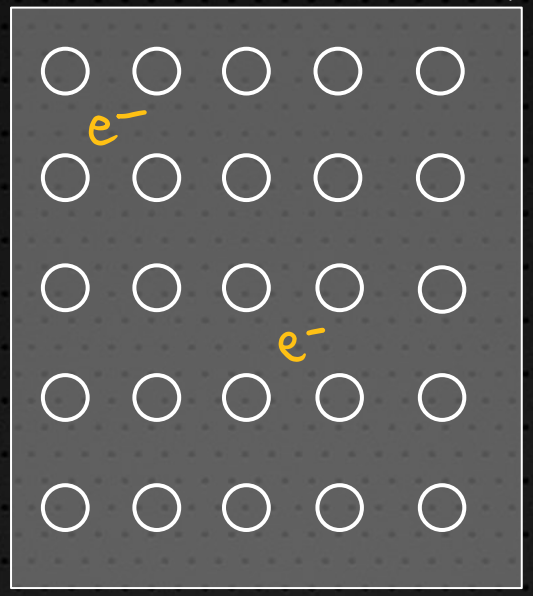
1. FUNDAMENTALS

○ → holes

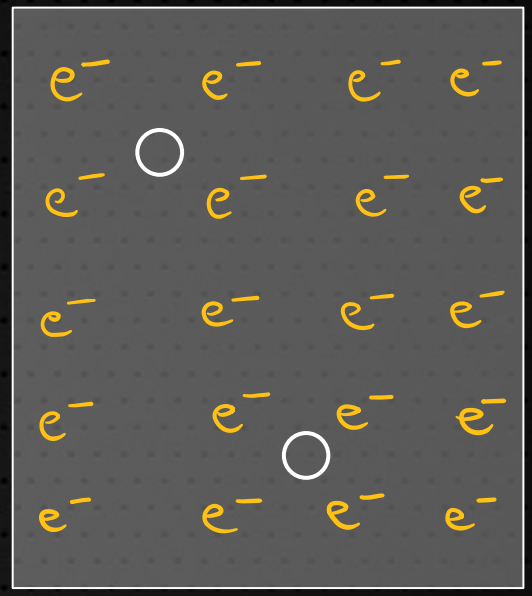
e^- → conduction electron

Extrinsic semiconductor →

P-TYPE (Trivalent impurity)



n-TYPE (Pentavalent impurit)



Majority charge carriers — holes — electrons

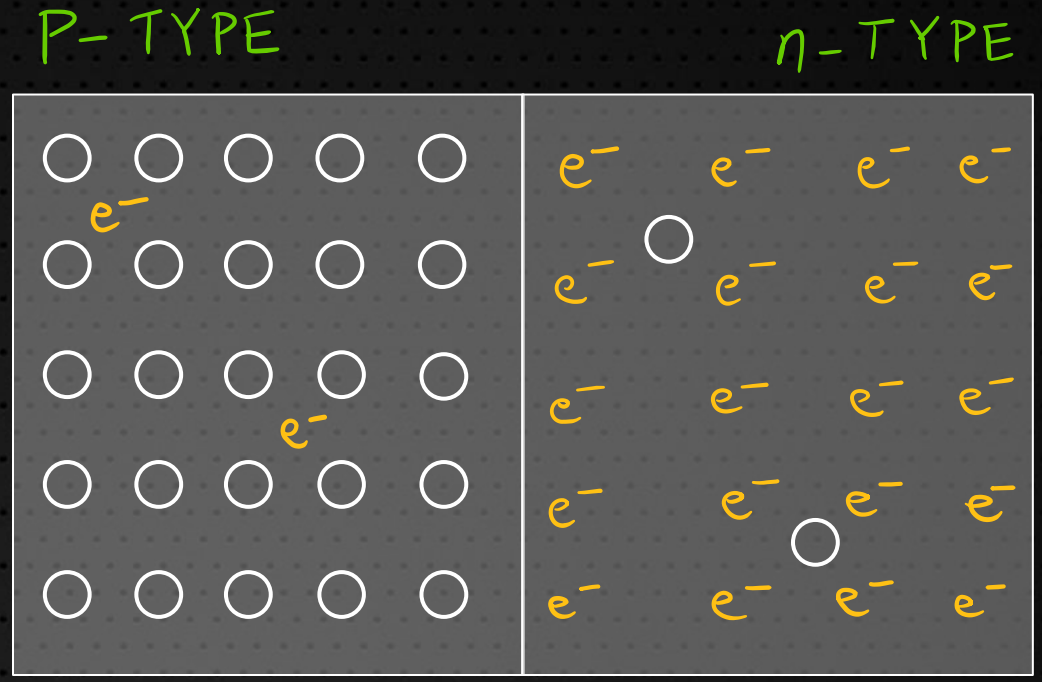
Minority charge carriers — electrons — holes

Eduniti for Physics

1. FUNDAMENTALS

○ ⟶ holes

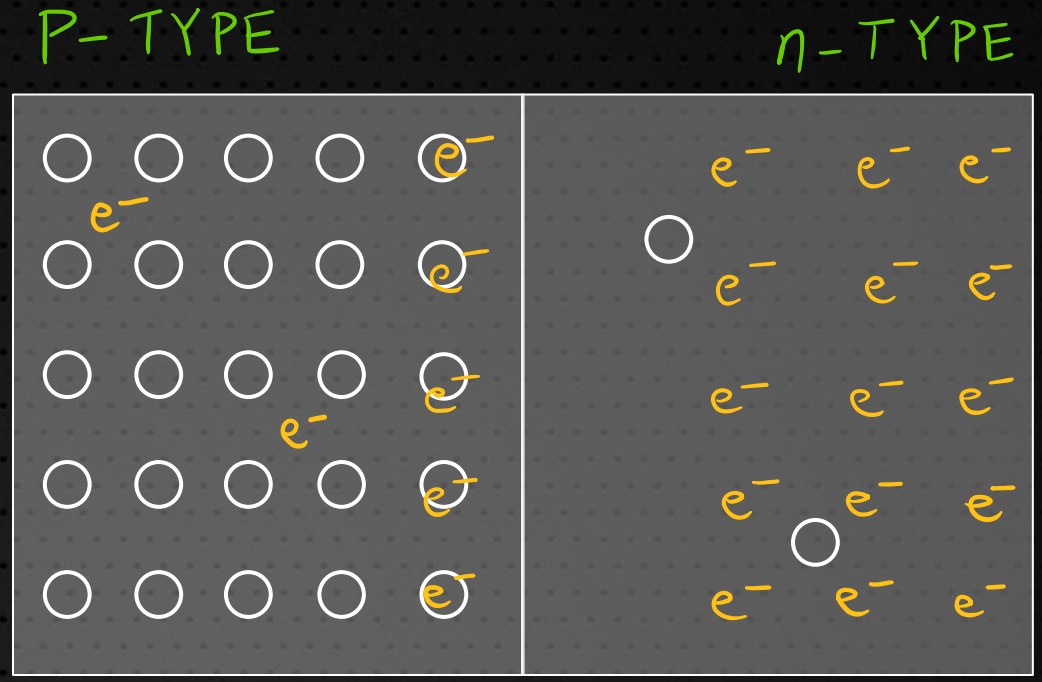
e^- ⟶ conduction electron



1. FUNDAMENTALS

○ ⟶ holes

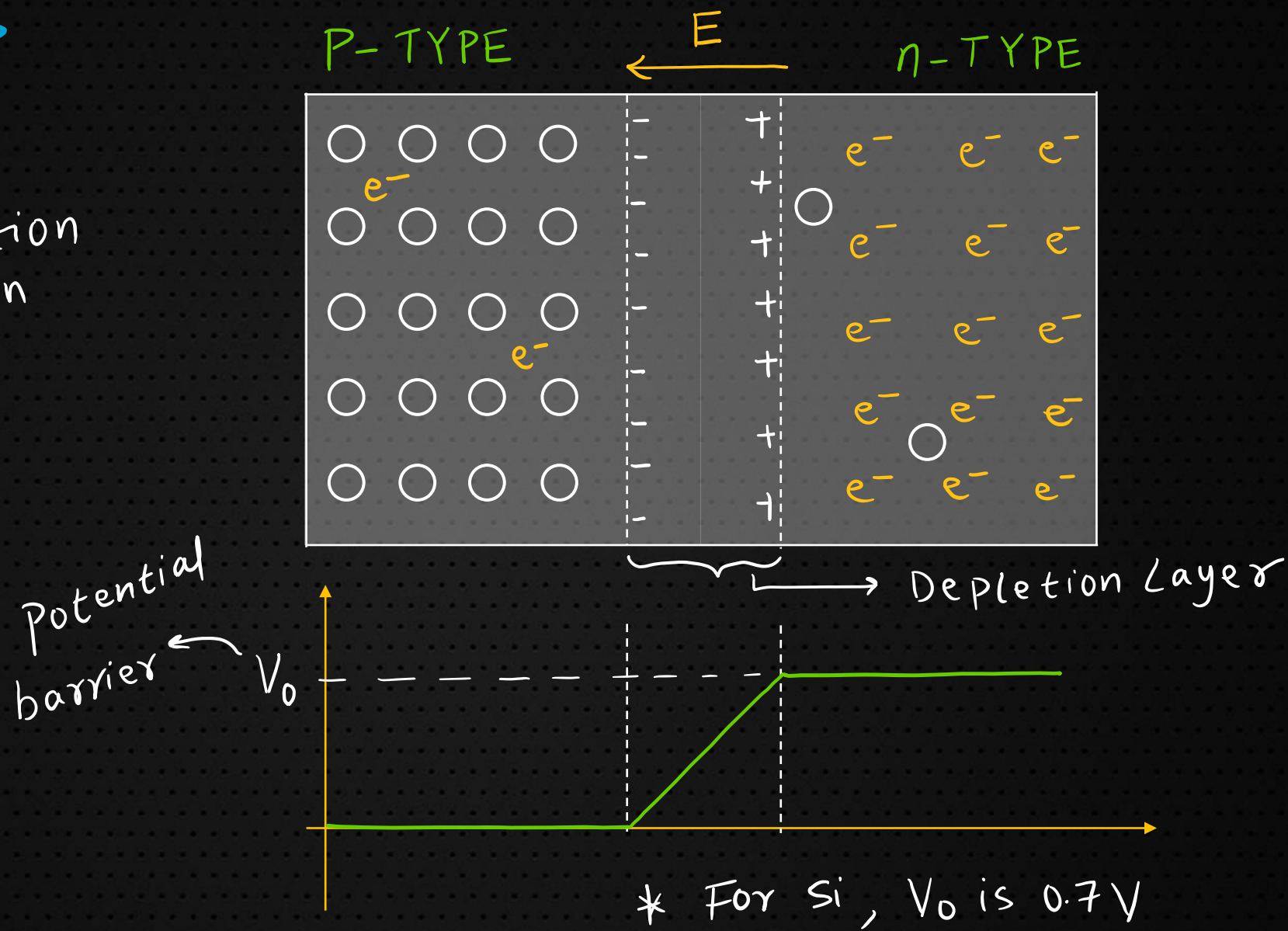
e^- ⟶ conduction electron



1. FUNDAMENTALS

○ → holes

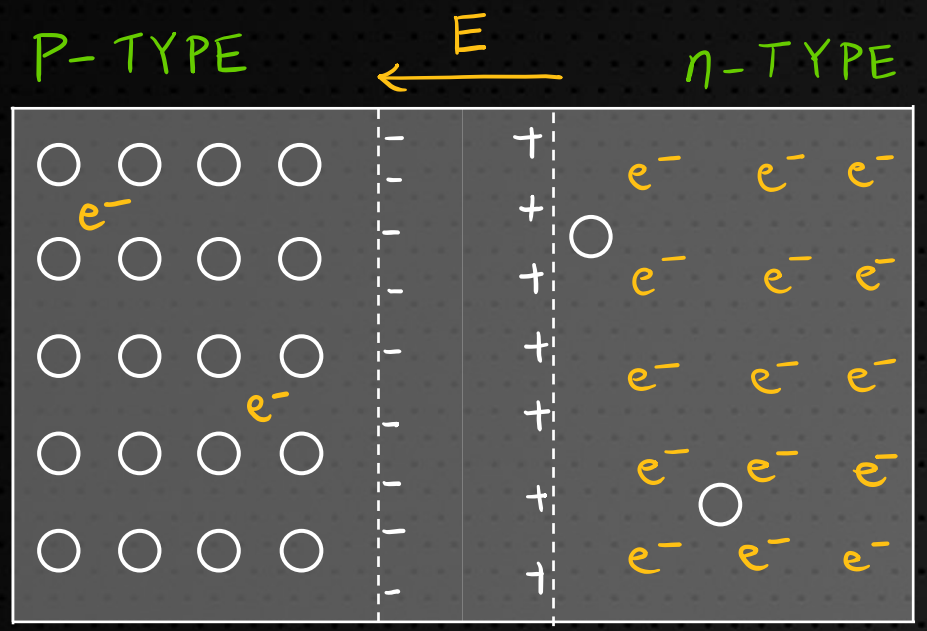
e^- → conduction electron

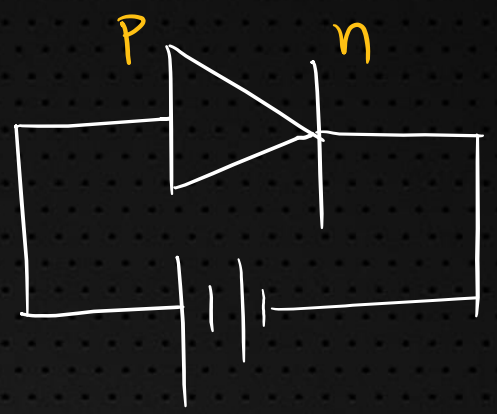


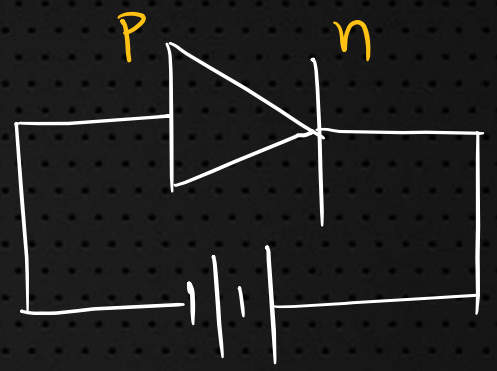
* For Si, V_0 is 0.7 V

EDUNITI

2. FORWARD AND REVERSE BIASED







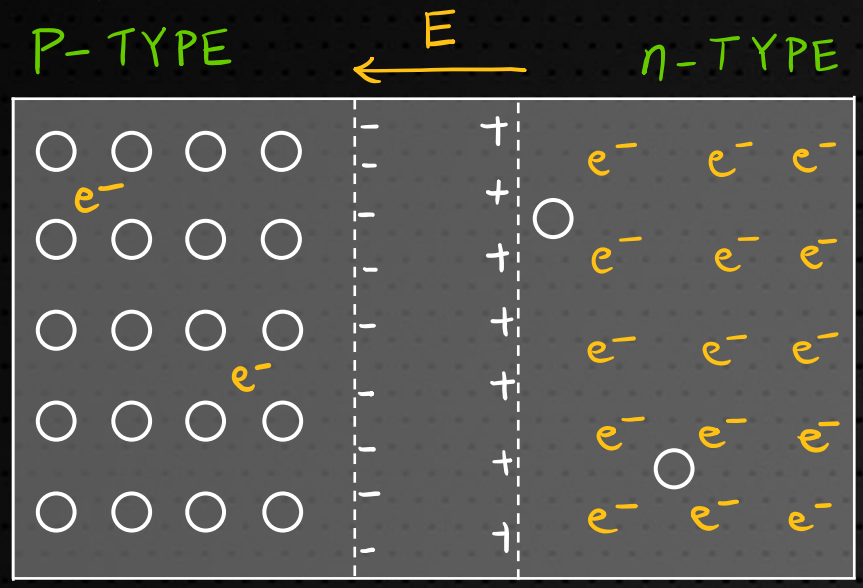
Forward biased
↳ current due to majority charge carriers.

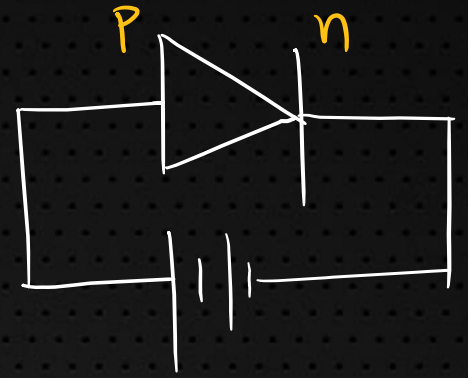
Reverse biase
↳ very small current due to minority charge carriers.

Eduniti for Physics

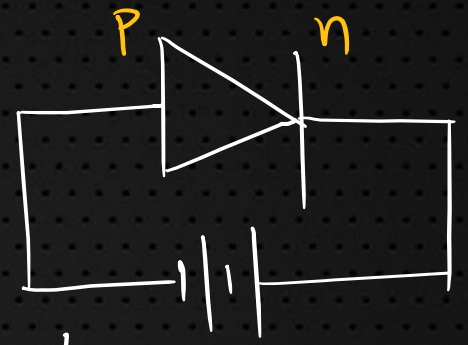
EDUNITI

2. FORWARD AND REVERSE BIASED

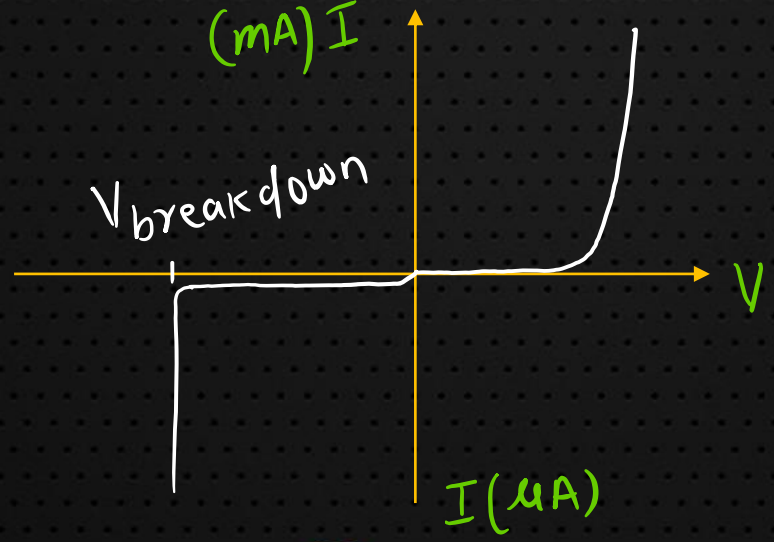




Forward biased
↳ current due to majority charge carriers.



Reverse biase
↳ very small current due to minority charge carriers.



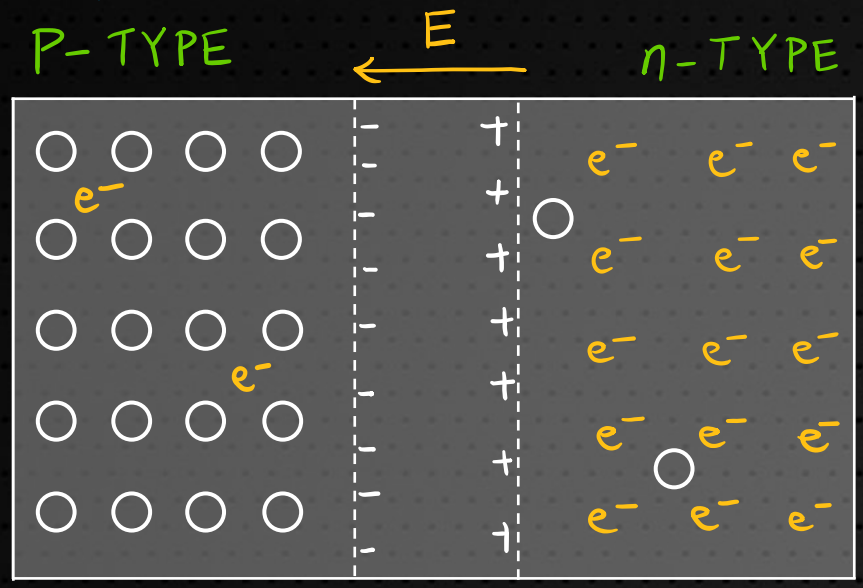
①

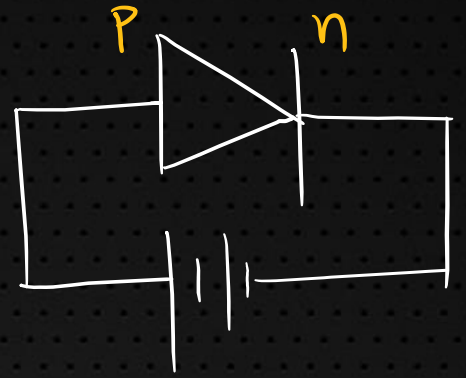
For ideal diode Resistance is zero (FB) and infinite (RB).

Eduniti for Physics

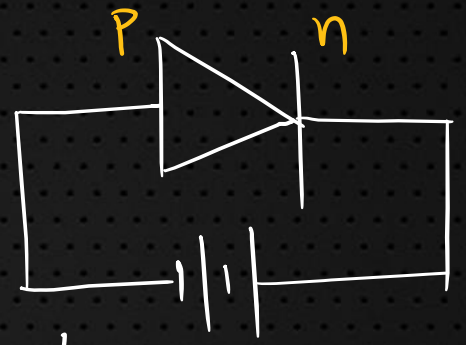
EDUNITI

2. FORWARD AND REVERSE BIASED

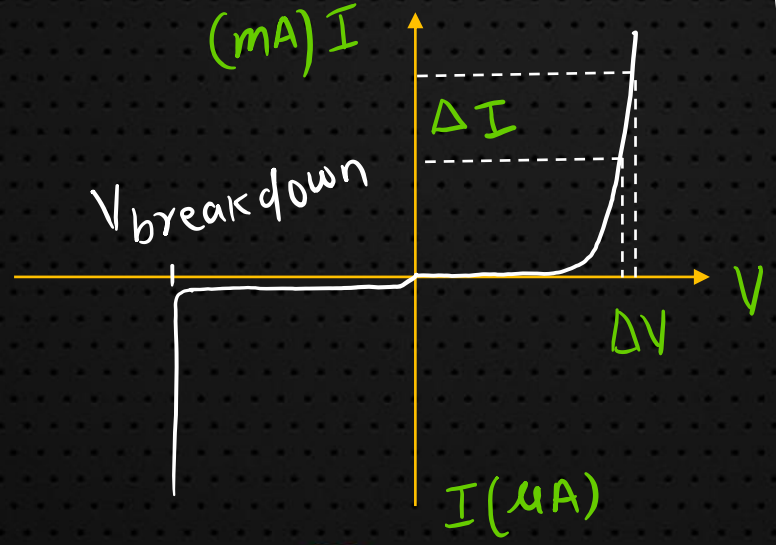




Forward biased
↳ current due to majority charge carriers.



Reverse biase
↳ very small current due to minority charge carriers.



②

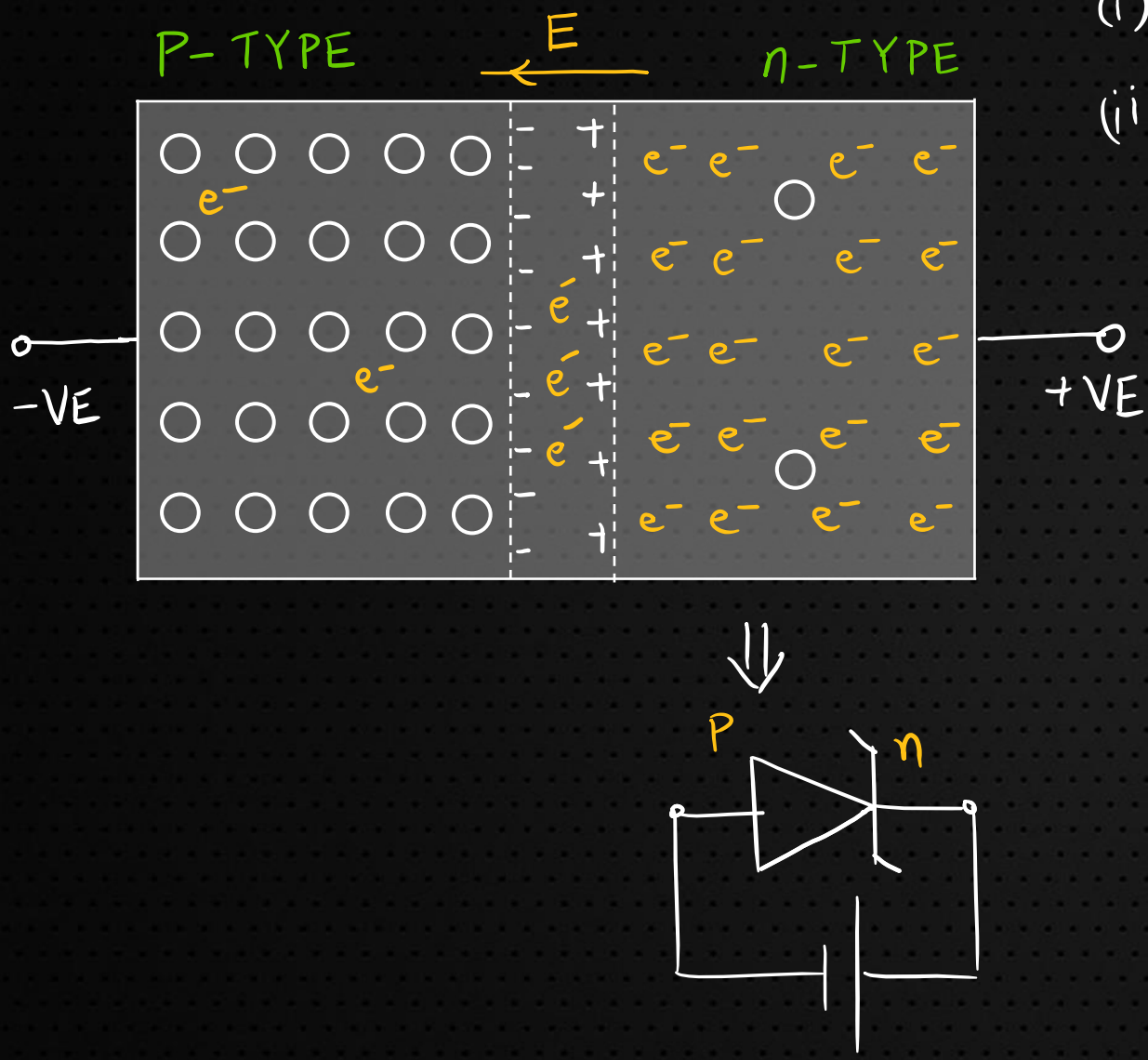
Dynamic Resistance,

$$R = \frac{\Delta V}{\Delta I}$$

Eduniti for Physics

EDUNITI

3. ZENER DIODE



(i) It is heavily doped

(ii) Depletion region very thin
⇒ E is very large.

(iii) In RB state, E strength ↑ and e^- covalent bonds in depletion region breaks.

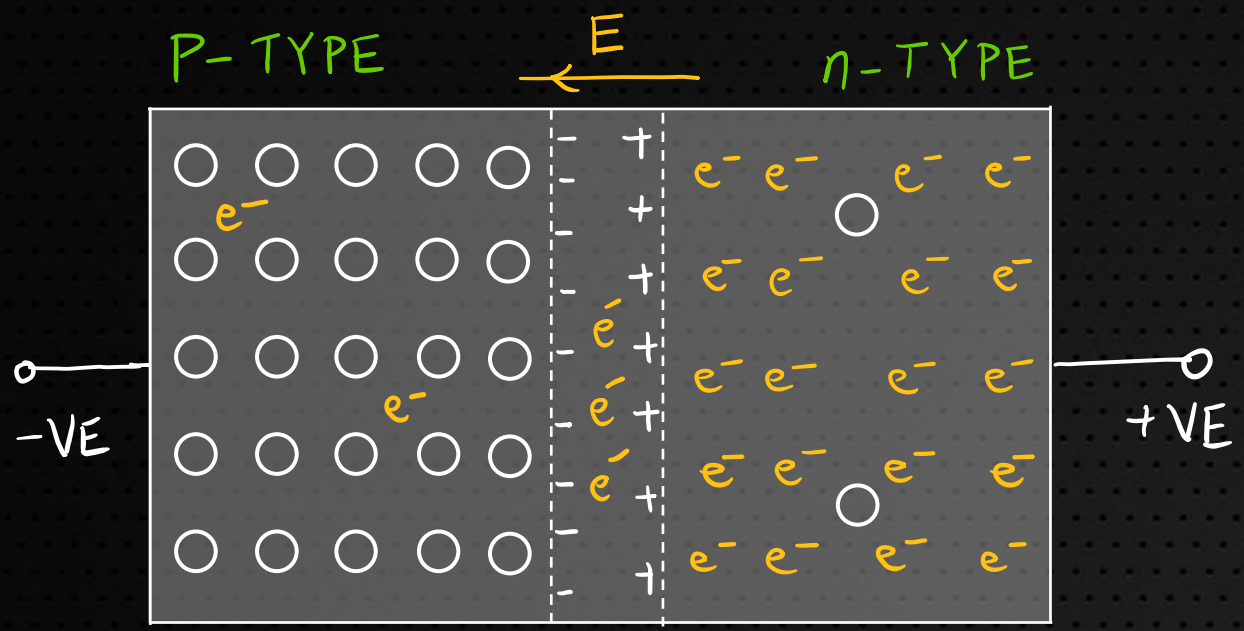
(iv) This e^- moves to n side and reverse current flows from n to P.

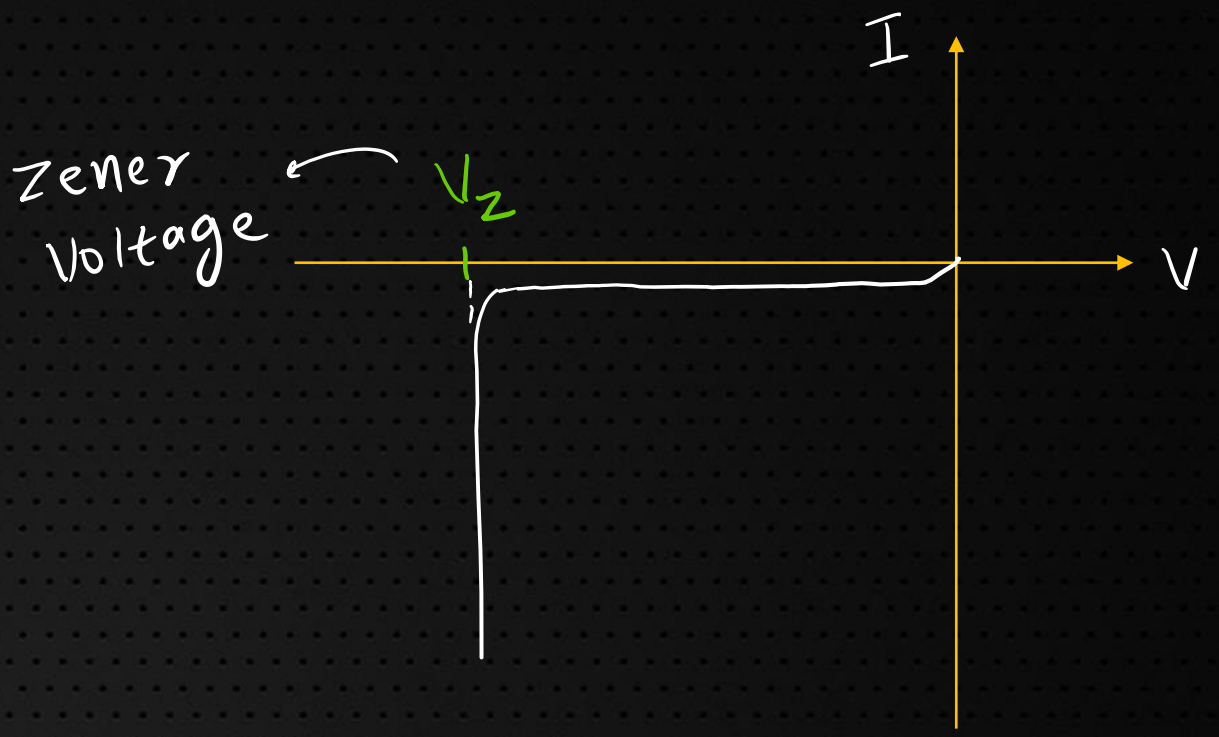
Zener Breakdown

Eduniti for Physics

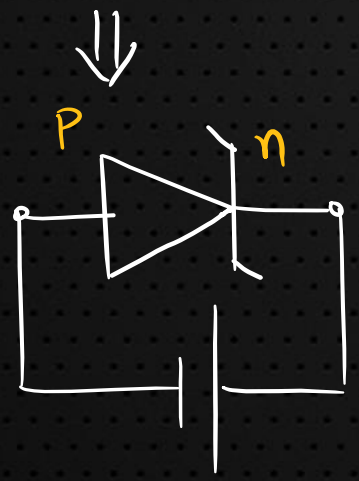
EDUNITI

3. ZENER DIODE





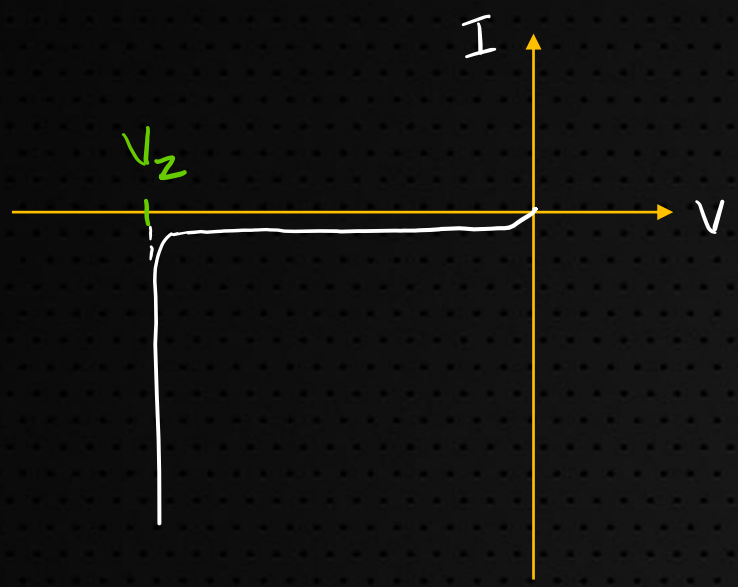
USED AS Voltage Regulator.

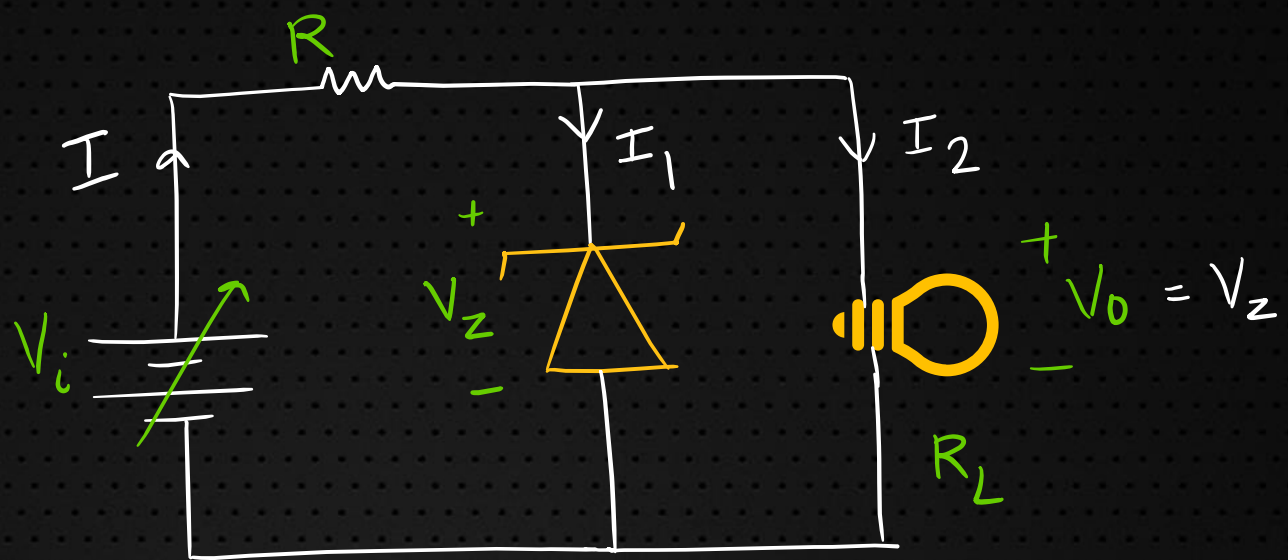


Eduniti for Physics

EDUNITI

3. ZENER DIODE





(1) ↳ As p.d. across crosses V_z (breakdown Voltage), almost all current passes through diode.

(2) ↳ And constant V_z is across it

(3) $I_2 = \frac{V_z}{R_L}$ (4) $I = I_1 + I_2$

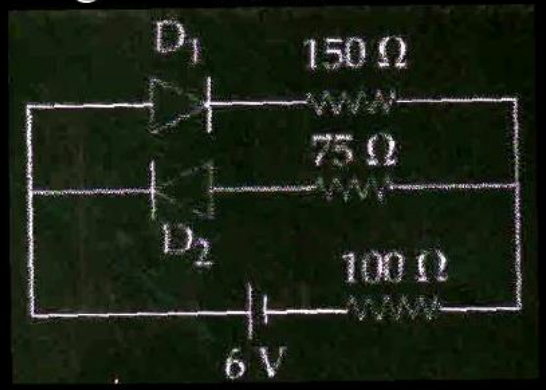
Eduniti for Physics

EDUNITI

5 PYQs For UNDERSTANDING

Eduniti for Physics

1. The circuit shown below contains two ideal diodes, each with a forward resistance of 50Ω. If the batterty voltage is 6V, the current through the 100Ω resistance (in Amperes) is

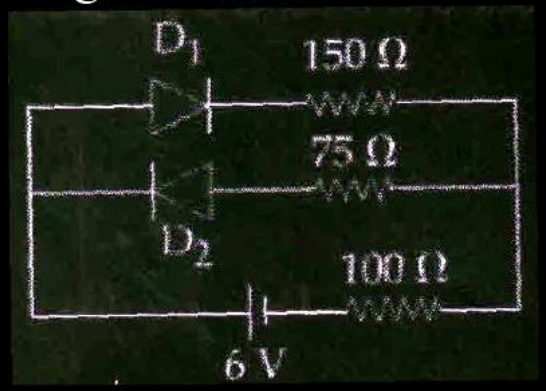


(a) 0.036
(b) 0.020
(c) 0.027
(d) 0.030

JEE Main 2019

Solution on Next Page

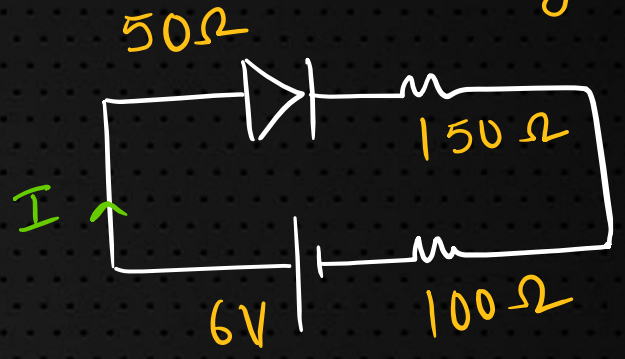
1. The circuit shown below contains two ideal diodes, each with a forward resistance of 50Ω. If the batterty voltage is 6V, the current through the 100Ω resistance (in Amperes) is



(a) 0.036
(b) 0.020
(c) 0.027
(d) 0.030

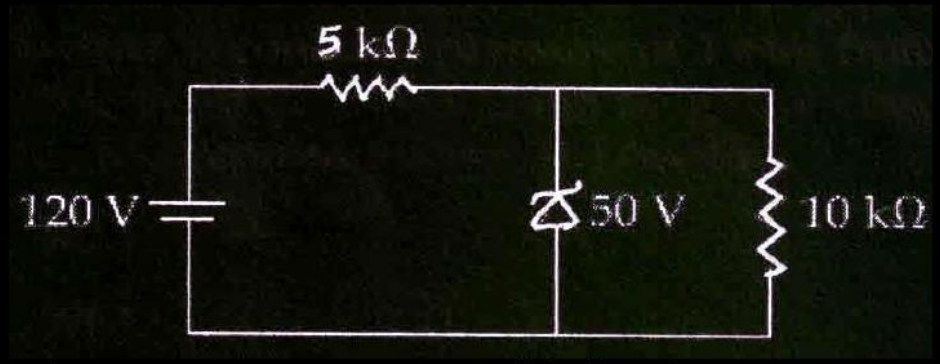
JEE Main 2019

Soln: Ideal diode in reverse state offers infinite resistance. ∴ no current through D_2



$$I = \frac{6}{5+150+100} = \boxed{0.02\ A}$$

2. For the circuit shown below, the current through the Zener diode is

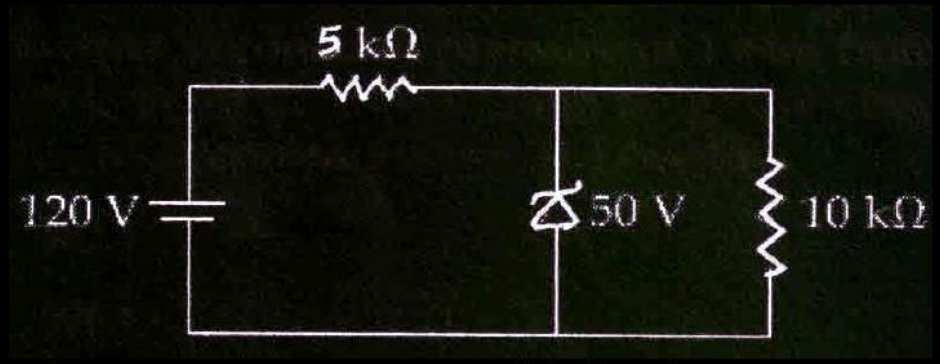


(a) 9 mA
(b) 5 mA
(c) Zero
(d) 14 mA

JEE Main 2019

Solution on Next Page

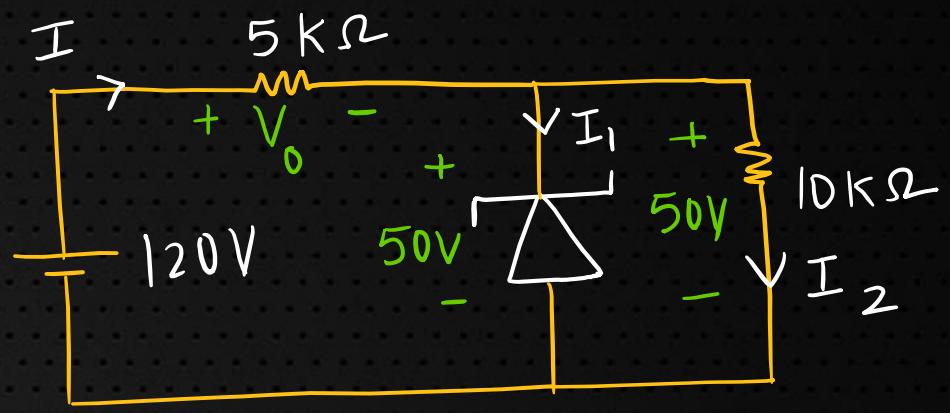
2. For the circuit shown below, the current through the Zener diode is



(a) 9 mA
(b) 5 mA
(c) Zero
(d) 14 mA

JEE Main 2019

Sol^{n}: Zener diode is in Breakdown mode.



$$I_2 = \frac{50\ V}{10\ K\Omega} = 5\ mA$$

$$V_0 = 120 - 50 = 70\ V$$

$$\therefore I = \frac{70\ V}{5\ K\Omega} = 14\ mA$$

$$\Rightarrow I_1 = I - I_2 = 14 - 5 = \boxed{9\ mA}$$

3. In the given circuit, the current through Zener Diode is close to

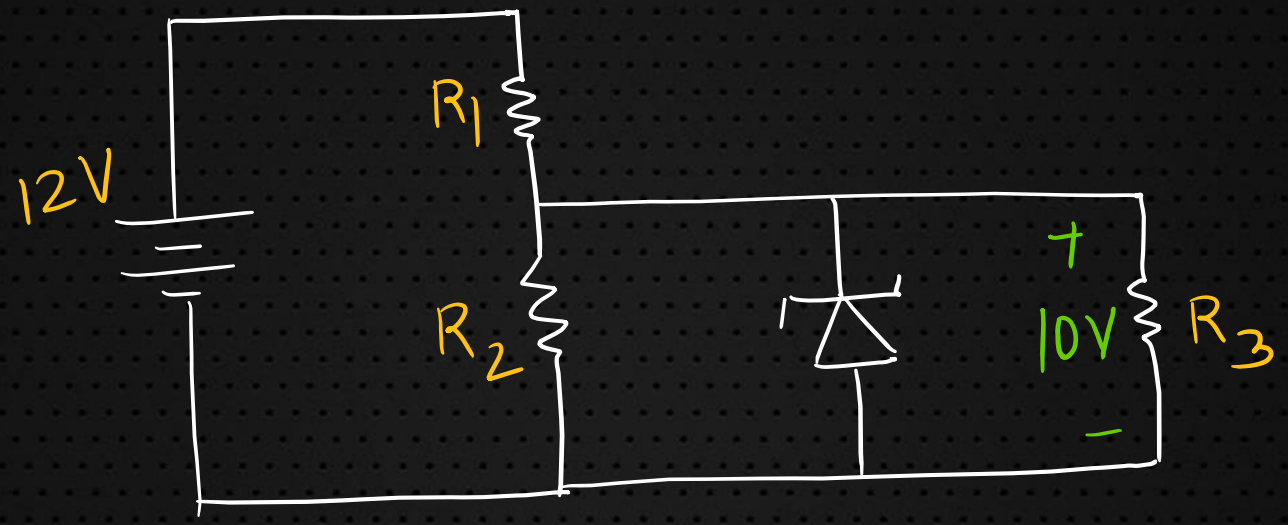
(a) 0.0 mA (b) 6.7 mA

(c) 4.0 mA (d) 6.0 mA

JEE Main 2019

$R_1 = 500\,\Omega$

$R_2 = 1500\,\Omega$



Solution on Next Page

EDUNITI

3. In the given circuit, the current through Zener Diode is close to

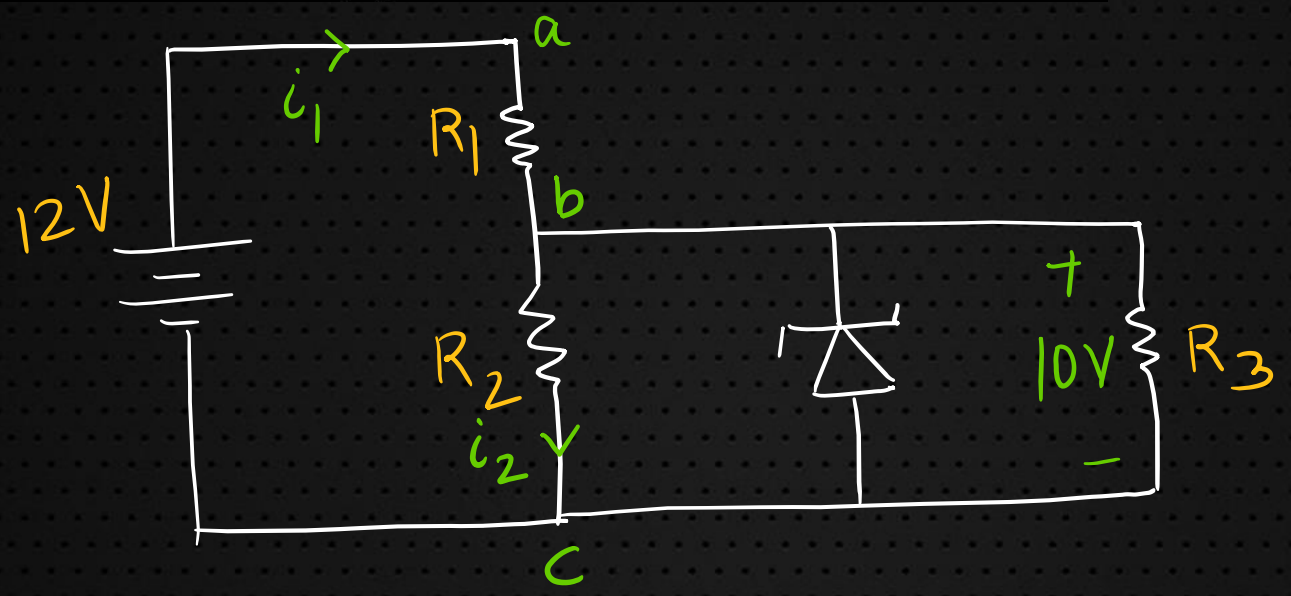
(a) 0.0 mA (b) 6.7 mA

(c) 4.0 mA (d) 6.0 mA

JEE Main 2019

$R_1 = 500\,\Omega$

$R_2 = 1500\,\Omega$



Soln°: Assume Zener in Breakdown mode.

$\therefore V_{bc} = 10V$ and $V_{ab} = 2V$

$i_1 = \frac{2}{500} = 4\,mA$

$i_2 = \frac{10}{1500} = 6.7\,mA$

$\because i_1 < i_2$ not possible

Hence Zener is not in Breakdown and Current through it is **Zero.**

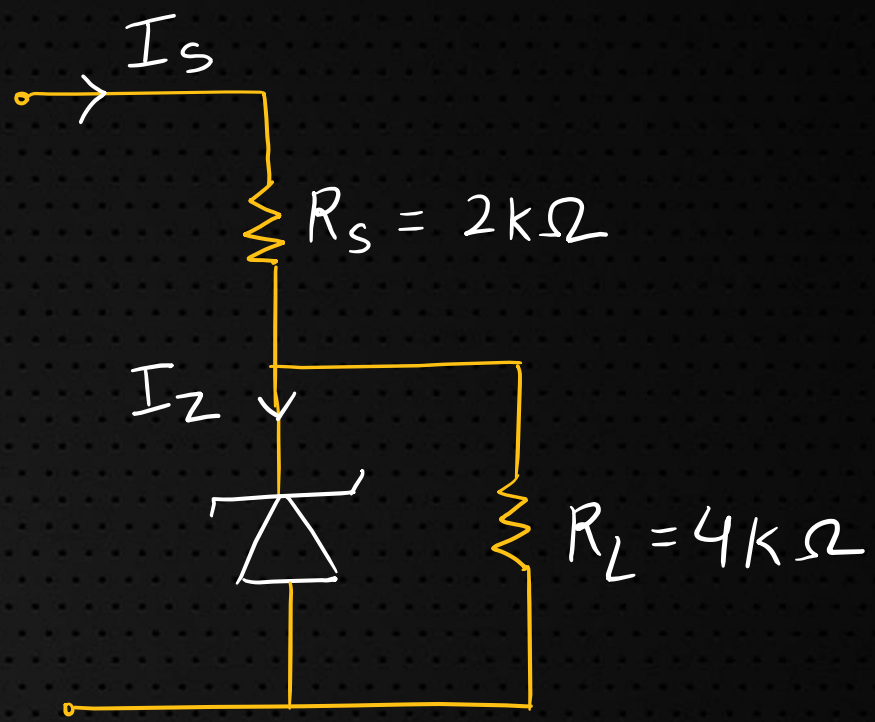
Eduniti for Physics

4. Figure shows a *DC* voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is the maximum Zener current?

(a) 2.5 mA (b) 1.5 mA

(c) 7.5 mA **(d)** 3.5 mA

JEE Main 2019



Solution on Next Page

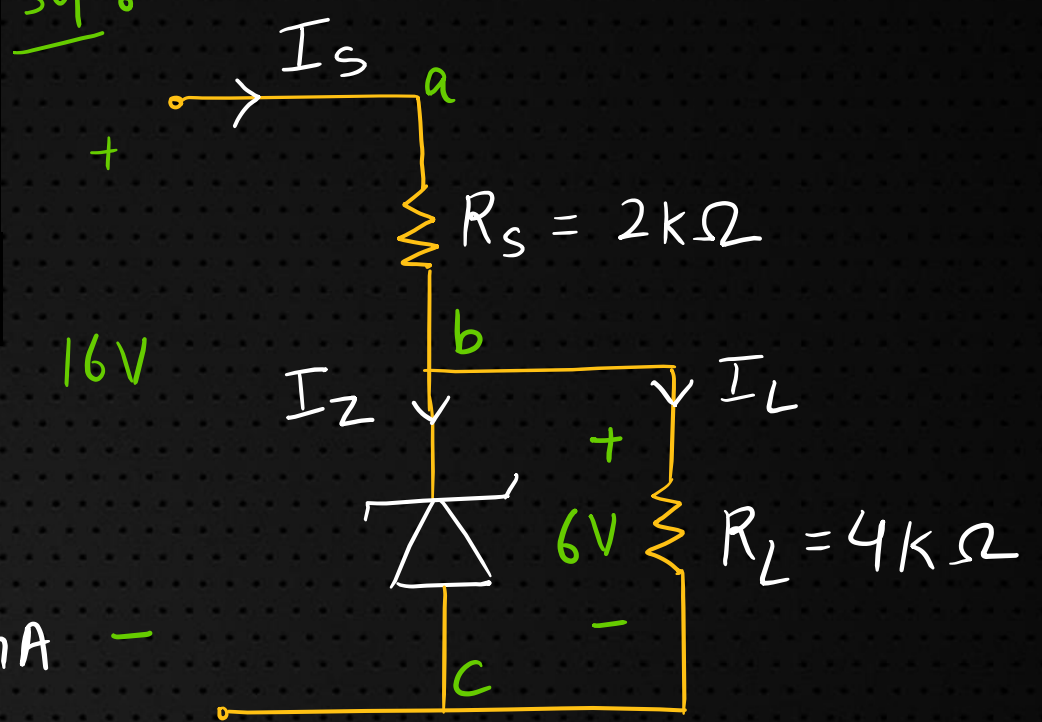
4. Figure shows a *DC* voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is the maximum Zener current?

(a) 2.5 mA (b) 1.5 mA

(c) 7.5 mA (d) 3.5 mA

JEE Main 2019

soln:



I_z is max when $V_{in} = 16V$

also zener will be in Breakdown mode

$$\therefore I_L = \frac{6V}{4K\Omega} = 1.5mA$$

$$V_{ab} = 16 - 6 = 10V$$

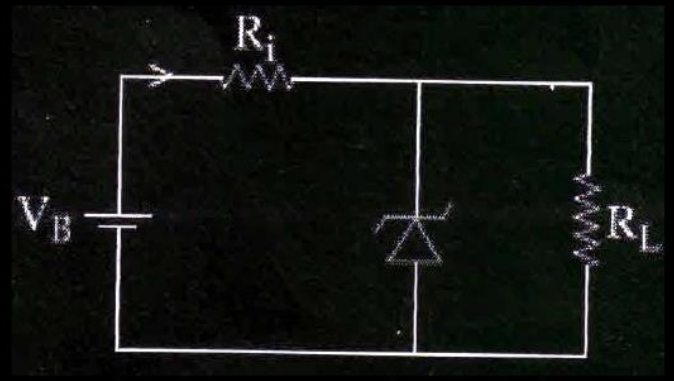
$$\therefore I_s = \frac{10V}{2K\Omega} = 5mA$$

$$\Rightarrow I_z = I_s - I_L$$

$$= 5 - 1.5 = \boxed{3.5mA}$$

Solution on Next Page

5. The figure represents a voltage regulator circuit using a Zener diode. The breakdown voltage of the Zener diode is 6 V and the load resistance is, $R_L = 4\ \text{k}\Omega$. The series resistance of the circuit is $R_i = 1\ \text{k}\Omega$. If the battery voltage V_B varies from 8 V to 16 V, what are the minimum and maximum values of the current through Zener diode?



(a) 0.5 mA; 6 mA
(b) 1 mA; 8.5 mA
(c) 0.5 mA; 8.5 mA
(d) 1.5 mA; 8.5 mA

JEE Main 2019

Soln: $I_2 = \frac{6\,V}{4\,k\Omega} = 1.5\,mA$

For Min: $V_B = 8\,V$, $V_1 = 8 - 6 = 2\,V$

$\therefore I = \frac{2\,V}{1\,k\Omega} = 2\,mA$

$\Rightarrow I_1 = I - I_2 =$ **0.5 mA**

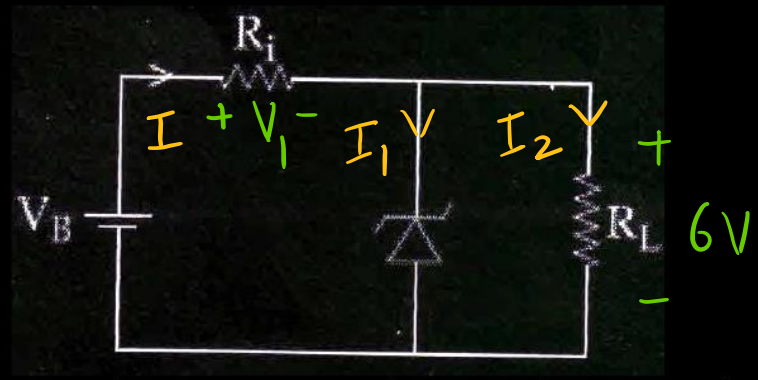
For Max: $V_B = 16\,V$, $V_1 = 16 - 6 = 10\,V$

$\therefore I = \frac{10\,V}{1\,k\Omega} = 10\,mA$

$\Rightarrow I_1 = I - I_2$

$= 8.5\,mA$

5. The figure represents a voltage regulator circuit using a Zener diode. The breakdown voltage of the Zener diode is 6 V and the load resistance is, $R_L = 4\,k\Omega$. The series resistance of the circuit is $R_i = 1\,k\Omega$. If the battery voltage V_B varies from 8 V to 16 V, what are the minimum and maximum values of the current through Zener diode?



(a) 0.5 mA; 6 mA
(b) 1 mA; 8.5 mA
(c) 0.5 mA; 8.5 mA
(d) 1.5 mA; 8.5 mA

JEE Main 2019